Atlas AI — Complete Operations & Architecture Guide

Author: Andrew H. Bond **Last Updated:** 2026-04-04 **Status:** Production (systemd-managed services) **URL:** https://atlas-sjsu.duckdns.org

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1. Hardware

HP Z840 Workstation ("Atlas")

Component	Specification
CPU	2x Intel Xeon E5-2690 v3 (24 cores / 48 threads, 2.60 GHz)
RAM	252 GB DDR4 (mixed DIMMs)
GPU 0	NVIDIA Quadro GV100 32 GB HBM2 (Volta, compute 7.0, bus 04:00.0, CPU 0)
GPU 1	NVIDIA Quadro GV100 32 GB HBM2 (Volta, compute 7.0, bus 84:00.0, CPU 1)
NVLink	Not possible (one x16 root port per CPU)
SSD	1.8 TB NVMe (48% used)
Archive	15 TB RAID5 (/archive, 2% used)
Network	Tailscale VPN (100.68.134.21), LAN 192.168.0.7
Location	Bel Marin Keys, Novato, CA

Thermal Limits

- CPU package: crit at 100°C, cap joblib workers at 20 (40 causes 99°C)
- GPU 0 (Spock): typically 57°C idle, 80°C under load
- GPU 1 (Kirk/embed): typically 78-83°C under load

Software Stack

Component	Version
OS	Ubuntu 24.04.2 LTS (kernel 6.8.0-106)
NVIDIA Driver	570.211.01
CUDA	12.8
Python	3.12.3 (venv at /home/claude/env)
PyTorch	2.10.0+cu128
PostgreSQL	16 with pgvector 0.6.0
NATS	v2.10.24 (JetStream enabled)

Component	Version
llama.cpp	Latest (built from source)
Caddy	v2.x (HTTPS + TLS via Let's Encrypt)
CuPy	cu128 (for CUDA RawKernels)

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2. Architecture

2.1 Cognitive Architecture Overview

Atlas implements a 10-subsystem cognitive architecture inspired by dual-process theory (Kahneman, 2011), Global Workspace Theory (Baars, 1988), and biologically-inspired memory hierarchies.

```
┌──────────────────────────┐
│   Web UI (atlas-chat)    │
└──────────────────────────┘
            │ HTTPS
            ▼
┌──────────────────────────┐
│  Caddy (TLS + Reverse)   │ :443
└──────────────────────────┘
            ▼
┌──────────────────────────┐
│  OAuth2 Proxy (Google)   │ :4180
└──────────────────────────┘
            ▼
┌──────────────────────────┐
│    RAG Server (Flask)    │ :8081
│  Wiki → PCA-384 → FTS    │
└──────────────────────────┘
            ▼                ▼
┌──────────────────┐  ┌──────────────────┐
│ Spock (GPU 0)    │  │  Kirk (GPU 1)    │
│ Qwen 72B Q5      │  │ (when available) │
│ :8080            │  │  :8082           │
└──────────────────┘  └──────────────────┘
            ▼
┌──────────────────────────────────────┐
│      NATS JetStream :4222            │
│  Event Fabric (agi.* subjects)       │
└──────────────────────────────────────┘
   │     │     │     │     │     │
Memory Safety Meta DHT  Env  Integration
:50300 :50055       ...
```

2.2 Dual-Hemisphere Design

Based on Iain McGilchrist's "The Master and His Emissary" (2009) and dual-process theory:

- **Left Hemisphere (Spock)**: Analytical, precise, citation-heavy. Gemma 4 or Qwen 72B. Excels at code analysis, formal reasoning, factual Q&A.

- **Right Hemisphere (Kirk)**: Creative, pattern-matching, divergent. Qwen 32B. Excels at brainstorming, cross-domain connections, narrative.
- **Debate Protocol**: 4-round structured debate with parallel evaluation, inspired by "Debate" (Irving et al., 2018) and "Constitutional AI" (Bai et al., 2022).
- **Confidence Metric**: Hemisphere disagreement measured via embedding cosine distance, calibrated via Platt scaling.

2.3 Memory Hierarchy

Inspired by Atkinson-Shiffrin (1968) multi-store model and Tulving's (1972) memory taxonomy:

Tier	Medium	Latency	Contents	Compression
L1	VRAM (KV cache)	<1 ms	Current conversation	fp16 (q8_0 available)
L2	RAM	~1 ms	Hot embeddings, recent sessions	TQ3 optional
L3	SSD (PostgreSQL)	~5 ms	Semantic + episodic + procedural	PCA-384 + IVFFlat
L4	HDD (RAID5)	~50 ms	Full repos, archives	PCA-384 + TQ3 (27.7x)
L5	Network	~100 ms+	GitHub, web, BitTorrent	NATS codec (89.5x)

Memory types (Tulving, 1972): - **Semantic**: Factual knowledge, RAG chunks. pgvector with BGE-M3 1024-dim. - **Episodic**: Conversation history. Timestamped episodes with embeddings. - **Procedural**: Learned operations. SQLite with trigger-pattern matching.

2.4 Safety Architecture (ErisML DEME)

Three-layer ethical firewall based on the DEME (Distributed Ethical Moral Engine) framework, grounded in cross-civilizational moral philosophy:

Layer	Latency	Function	Theoretical Basis
Reflex	<1 ms	PII detection, prompt injection, content policy	Kahneman System 1
Tactical	~100 ms	MoralVector assessment via Ethics Modules	Virtue ethics (Aristotle), Deontology (Kant)
Strategic	~1 s	SHA-256 hash-chained decision	Consequentialism (Mill), Social contract (Rawls)

Layer	Latency	Function	Theoretical Basis
		proofs, audit trail	

Ethics corpus: 2.4M passages from 9 traditions spanning 3,300+ years: Confucian, Buddhist, Hindu, Islamic, Christian, Jewish, Greco-Roman, African, Indigenous.

2.5 Search Architecture (Benchmark-Proven)

Three-tier cascade validated by empirical benchmarks (50 queries, 112K chunks):

Tier	Method	Recall@10	Latency	When Used
1	Wiki article lookup	—	<1 ms	Structural "how does X work?" queries
2	**PCA-384 IVFFlat**	**0.906**	**4.4 ms**	**Primary search (all queries)**
3	tsvector FTS	0.102	2.9 ms	Keyword fallback (vector returns 0)

Methods tested but NOT deployed (benchmarks showed inferior results):

Method	Recall@10	Latency	Why Not
Full 1024-dim exact	1.000	457 ms	104x slower than PCA-384
GPU Hamming funnel	0.908	10.8 ms	Same recall, 2.5x slower at 112K
Hybrid RRF (vector+FTS)	0.852	14.9 ms	FTS noise dilutes vector results

Key finding: RRF fusion of vector + FTS HURTS recall. FTS is useful only as a standalone keyword fallback, not as a fusion source.

References: - PCA-Matryoshka: Bond (IEEE TAI, 2026) - TurboQuant: Zandieh, Han, Daliri, Karbasi (ICLR, 2026) - Reciprocal Rank Fusion: Cormack, Clarke, Butt (SIGIR, 2009) - BGE-M3: Xiao et al. (ACL Findings, 2024) - IVFFlat: Jégou, Douze, Schmid (IEEE TPAMI, 2011)

2.6 Compression Pipeline (PCA-Matryoshka + TurboQuant)

Theory: PCA rotation reorders embedding dimensions by explained variance, enabling effective truncation for models not trained with Matryoshka losses (Kusupati et al., NeurIPS 2022). Combined with Lloyd-Max scalar quantization on the rotated hypersphere (Zandieh et al., ICLR 2026).

Pipeline: Embed → Center → PCA Rotate → Truncate (384 dims) → Quantize (3-bit) → Bit-pack

Method	Cosine	Bytes/vec	Ratio	Index Size
Full 1024 float32	1.000	4,096 B	1.0x	24 GB (ethics)
PCA-384 float32	0.984	1,536 B	2.7x	3.7 GB
TQ3 1024-dim	0.978	388 B	10.6x	—
PCA-384 + TQ3	0.971	148 B	27.7x	—
Binary PCA-384	—	48 B	85x	5.25 MB
NATS JSON payload	—	254 B	89.5x	—

KV cache compression (Volta limitation): q8_0 KV saves 47% memory but incurs 1.9x throughput penalty on Volta (no hw int8). fp16 retained for production. Enable `--cache-type-k q8_0 --cache-type-v q8_0` on Ampere+ GPUs.

2.7 Wiki Compilation (Karpathy Pattern)

Inspired by Andrej Karpathy's "LLM Knowledge Bases" approach: - Spock (Gemma 4) reads source code and writes structured markdown articles - Articles have backlinks, summaries, and key concept indexes - Wiki is Tier 1 in search cascade (instant lookup, no embedding needed) - Self-healing: periodic lint passes detect stale/inconsistent articles - Location: `/home/claude/agi-hpc/wiki/`

2.8 Models & Personas

Spock (Left Hemisphere — GPU 0)

Property	Value
Persona	Analytical, precise, citation-heavy. Named after the Vulcan science officer.
Role	Primary reasoning engine. Handles factual Q&A, code analysis, formal logic.
Current Model	Qwen2.5-72B-Instruct-Q5_K_M (GGUF, 5-bit quantized)
VRAM	~31 GB on GPU 0
Context	8,192 tokens (14,336 with q8_0 KV cache, at 2x speed penalty)
Throughput	~4.2 tok/s (fp16 KV), ~2.2 tok/s (q8_0 KV)
Port	8080
System prompt	"You are Atlas... the Left Hemisphere — analytical, precise, and citation-heavy."

Model history: Originally Gemma 4 31B-IT Q5_K_M. Upgraded to Qwen2.5-72B for superior instruction following and reasoning. The `start_atlas.sh` script references both; the systemd service uses whichever GGUF is in the models directory.

Kirk (Right Hemisphere — GPU 1)

Property	Value
Persona	Creative, intuitive, pattern-matching. Named after the Enterprise captain.
Role	Divergent thinking. Brainstorming, cross-domain analogy, narrative synthesis.
Planned Model	Qwen3-32B-Q5_K_M or equivalent creative model
VRAM	~31 GB on GPU 1 (when not running embedding jobs)
Context	8,192 tokens
Port	8082
System prompt	"You are Atlas... the Right Hemisphere — creative, intuitive, pattern-seeking."

Status: Kirk is typically offline because GPU 1 is used for embedding jobs (BGE-M3 for ethics corpus, publications). When both hemispheres are active, the RAG server orchestrates a 4-round debate protocol.

Debate Protocol

When both Spock and Kirk are active:

1. **Opening** (parallel): Both receive the query + RAG context independently
2. **Challenge** (parallel): Each reads the other's response and critiques it
3. **Synthesis**: Kirk integrates both perspectives as "captain's decision"
4. **Confidence**: Measured from cosine distance between Spock/Kirk embeddings

The debate is inspired by Irving et al.'s "AI Safety via Debate" (2018), where adversarial questioning surfaces errors that a single model might miss.

Embedding Model

Property	Value
Model	BAAI/bge-m3 (via sentence-transformers)
Dimension	1024 (full), 384 (PCA-projected)

Property	Value
Device	CPU (in RAG server), GPU 1 (for batch embedding)
Normalization	L2-normalized for cosine similarity
PCA basis	Fitted on 10K chunk embeddings, 94.8% variance at 384 dims

Available Models on Atlas

Model	Size	Quant	Location	Use
Qwen2.5-72B-Instruct	45 GB	Q5_K_M	/home/claude/models/	Spock (primary)
Qwen2.5-32B-Instruct	22 GB	Q4_K_M	/home/claude/models/	Spock (fallback)
Qwen3-32B	22 GB	Q5_K_M	/home/claude/models/	Kirk
Gemma 4 31B-IT	21 GB	Q5_K_M	/home/claude/models/	Spock (original)
BAAI/bge-m3	2.5 GB	fp32	HuggingFace cache	Embedding

Persona Guidelines

Spock responses should: - Lead with the factual answer, then explain reasoning - Cite specific files, functions, or papers when referencing local knowledge - Use structured formatting (headers, bullet points, tables) - Flag uncertainty explicitly ("I don't have data on X")

Kirk responses should: - Start with the big-picture insight or analogy - Draw connections across different domains - Suggest creative alternatives the user may not have considered - Use narrative structure (story, metaphor)

Debate synthesis should: - Acknowledge where Spock and Kirk agree (convergent evidence) - Highlight where they disagree and why - Give the user actionable next steps - Report confidence level (high/medium/low based on disagreement metric)

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3. Services

3.1 systemd Services

All critical services are managed by systemd with auto-restart and boot persistence.

Service	Unit	Port	Process
Caddy	atlas-caddy	443, 80	HTTPS reverse proxy + TLS

Service	Unit	Port	Process
OAuth2 Proxy	atlas-oauth2-proxy	4180	Google authentication
RAG Server	atlas-rag-server	8081	Search + hemisphere routing
Spock LLM	atlas-llm-spock	8080	Qwen 72B Q5_K_M on GPU 0
Telemetry	atlas-telemetry	8085	System metrics + event stream

```
# Check all services
sudo systemctl status atlas-*

# Restart a service
sudo systemctl restart atlas-rag-server

# Follow logs
sudo journalctl -u atlas-rag-server -f

# Disable auto-start
sudo systemctl disable atlas-llm-spock

# Install/uninstall all
sudo bash deploy/systemd/install-services.sh
sudo bash deploy/systemd/install-services.sh --uninstall
```

3.2 Non-systemd Services (manual/tmux)

Service	Port	How to Start
NATS JetStream	4222	`nats-server -c /home/claude/nats.conf`
Research Portal	8443	Runs independently
Kirk LLM (GPU 1)	8082	Via start_atlas.sh when GPU 1 free
Memory NATS Service	50300	`python -m agi.memory.nats_service`
Safety NATS Service	50055	`python -m agi.safety.nats_service`
Metacognition	—	`python -m agi.metacognition.nats_service`

3.3 Request Flow

```
Browser → Caddy (:443, TLS)
  → OAuth2 Proxy (:4180, Google Auth)
    → RAG Server (:8081)
      → Wiki lookup (filesystem, <1ms)
      → PCA-384 IVFFlat search (pgvector, 4.4ms)
      → FTS fallback (tsvector, 2.9ms)
      → Embed query (BGE-M3, 232ms)
      → Inject context into prompt
      → Forward to Spock (:8080) or Kirk (:8082)
      → 4-round debate (if both hemispheres active)
      → Safety check
      → Stream response to client
```

```
Telemetry: Caddy routes /api/telemetry, /api/events, /api/visitors
  directly to Telemetry Server (:8085), bypassing OAuth2.
```

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4. Database

4.1 PostgreSQL Tables

Table	Rows	Size	Embeddings
ethics_chunks	2,391,361	~41 GB	1024-dim + PCA-384
chunks	112,001	~3.1 GB	1024-dim + PCA-384 + tsvector
publications	823,960	360 MB	Embedding in progress
episodes	9	<1 MB	1024-dim
cities	7,342	7 MB	PostGIS
countries	258	7 MB	PostGIS

4.2 Vector Indexes

Index	Type	Size	Scans	Notes
idx_ethics_embedding	IVFFlat	24 GB	6	Full 1024-dim (legacy)
idx_ethics_embedding_pca384	IVFFlat	3.7 GB	—	PCA-384, 1000 lists
idx_chunks_embedding	IVFFlat	1.67 GB	50	Full 1024-dim (legacy)
idx_chunks_embedding_pca384	IVFFlat	176 MB	active	PCA-384, 300 lists, probes=10
idx_chunks_tsv	GIN	—	—	Full-text search

4.3 PCA Model

- Path: `/home/claude/agi-hpc/data/pca_rotation_384.pkl`
- Fitted on: 10,000 sample embeddings from chunks table
- Variance captured: 94.8%
- Mean cosine similarity (round-trip): 0.985
- Format: pickle dict with `components` (384, 1024), `mean` (1024,), `explained_variance_ratio`

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5. Compression Summary

5.1 Storage Savings (Measured)

Component	Before	After	Savings
Ethics index	24 GB	3.7 GB	20.3 GB
Chunks index	1.67 GB	176 MB	1.5 GB
NATS payloads	22.7 KB/msg	254 B/msg	89.5x
Total index savings	**25.7 GB**	**3.9 GB**	**21.8 GB**

5.2 Throughput (Measured)

Operation	CPU	GPU (GV100)
PCA-384 projection	402K vec/s	10.5M vec/s (42x)
TQ3 compression	18.7K vec/s	—
Binary pack	—	included in Hamming

5.3 What Doesn't Work on Volta

Technique	Issue	Works On
KV cache q8_0/q4_0	1.9x throughput penalty (no hw int8)	Ampere+
Compressed-space search (Python)	1700x slower than float brute force	Needs C/CUDA kernel
Binary reranking (numpy)	4.5x slower than float matmul	Needs SIMD/CUDA
GPU ADC search	5x slower than CPU float (kernel overhead)	Needs optimization

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6. Key Files

Source Code

Path	Description
`src/agi/lh/`	Left Hemisphere (RAG, reasoning)
`src/agi/rh/`	Right Hemisphere (creative, pattern)
`src/agi/memory/`	Semantic, episodic, procedural memory
`src/agi/safety/`	DEME safety gateway
`src/agi/metacognition/`	Monitor, reflector, disagreement
`src/agi/core/events/`	NATS event fabric

Path	Description
`src/agi/common/hybrid_search.py`	3-tier search cascade, RRF, wiki
`src/agi/common/embedding_codec.py`	TQ3 NATS transport compression
`src/agi/common/embedding_service.py`	Shared BGE-M3 singleton
`src/agi/meta/llm/turboquant_kv.py`	KV cache compression

Operational Scripts

Path	Description
`scripts/start_atlas.sh`	Legacy tmux-based startup
`deploy/systemd/install-services.sh`	systemd service installer
`scripts/benchmark_search_tiers.py`	Regression test for search quality
`scripts/benchmark_tqpro_integration.py`	Compression quality benchmarks
`scripts/compile_wiki.py`	LLM wiki compilation (Karpathy pattern)
`scripts/migrate_ethics_pca384.py`	Ethics PCA-384 migration
`scripts/embed_publications.py`	Publications embedding job
`scripts/llm_adaptive_proxy.py`	Dynamic fast/long-context switching
`atlas-rag-server.py`	Main RAG server
`atlas-rag-indexer.py`	Repo indexer (auto PCA-384)

Configuration

Path	Description
`configs/fabric_config.yaml`	NATS subject hierarchy
`configs/lh_config.yaml`	Left Hemisphere settings
`configs/memory_config.yaml`	Memory service (BGE-M3, DB)
`configs/safety_config.yaml`	Safety thresholds
`/home/claude/.env`	OAuth2 secrets, API keys
`/home/claude/Caddyfile`	Reverse proxy routing

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7. Monitoring & Troubleshooting

Health Checks

```
# All services
sudo systemctl status atlas-*
```

```
# RAG server
curl http://localhost:8081/api/search-status

# LLM server
curl http://localhost:8080/health

# Telemetry
curl http://localhost:8085/api/telemetry

# GPU status
nvidia-smi --query-gpu=index,name,utilization.gpu,memory.used,temperature.gpu
 --format=csv

# CPU temps
sensors | grep Package
```

Common Issues

Symptom	Cause	Fix
Portal returns 502	oauth2-proxy not running	`sudo systemctl restart atlas-oauth2-proxy`
Search returns 0 results	tsvector not populated or IVFFlat probes=1	Run populate script; check probes=10
LLM very slow	KV cache quantized on Volta	Remove –cache-type-k/v flags
GPU OOM	Multiple models loaded	Check `nvidia-smi`, kill extra processes
CPU >95°C	Too many parallel workers	Cap at 20 threads (`taskset -c`)
PCA model missing	First deployment	Run `tqpro_migrate.py --fit-only`

Log Locations

Service	Log
RAG Server	`journalctl -u atlas-rag-server` or `/tmp/rag_server.log`
LLM Spock	`journalctl -u atlas-llm-spock` or `/tmp/qwen_server.log`

Service	Log
OAuth2	`journalctl -u atlas-oauth2-proxy`
Caddy	`journalctl -u atlas-caddy` or `/tmp/caddy.log`
Telemetry	`journalctl -u atlas-telemetry`

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8. Common Operations

Re-index Repositories

```
CUDA_VISIBLE_DEVICES=1 python3 atlas-rag-indexer.py --update
# Auto-computes embedding + PCA-384 for new/changed files
```

Rebuild PCA Model

```
python3 scripts/tqpro_migrate.py --fit-only
# Then restart RAG server to load new model
sudo systemctl restart atlas-rag-server
```

Compile Wiki

```
WIKI_DIR=/home/claude/agi-hpc/wiki python3 scripts/compile_wiki.py
# Lint existing articles
python3 scripts/compile_wiki.py --lint
```

Run Regression Benchmarks

```
CUDA_VISIBLE_DEVICES=1 python3 scripts/benchmark_search_tiers.py
# Quick smoke test
python3 scripts/benchmark_search_tiers.py --quick
```

Reload Hamming Index (after new chunks indexed)

```
curl -X POST http://localhost:8081/api/reload-index
```

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9. References

Core Architecture

1. Kahneman, D. “Thinking, Fast and Slow.” Farrar, Straus and Giroux, 2011.
2. Baars, B.J. “A Cognitive Theory of Consciousness.” Cambridge University Press, 1988.
3. McGilchrist, I. “The Master and His Emissary.” Yale University Press, 2009.
4. Tulving, E. “Episodic and Semantic Memory.” In Organization of Memory, 1972.

5. Atkinson, R.C. & Shiffrin, R.M. “Human Memory: A Proposed System.” Psychology of Learning and Motivation, 1968.

AI Safety

6. Irving, G., Christiano, P., Amodei, D. “AI Safety via Debate.” arXiv:1805.00899, 2018.
7. Bai, Y. et al. “Constitutional AI.” arXiv:2212.08073, 2022.

Embedding Compression

8. Bond, A.H. “PCA-Matryoshka: Enabling Effective Dimension Reduction for Non-Matryoshka Embedding Models.” IEEE TAI, 2026.
9. Zandieh, A., Han, I., Daliri, M., Karbasi, A. “Sub-linear Memory Inference via PolarQuant and QJL.” ICLR, 2026.
10. Kusupati, A. et al. “Matryoshka Representation Learning.” NeurIPS, 2022.
11. Xiao, S. et al. “BGE M3-Embedding.” ACL Findings, 2024.

Search & Retrieval

12. Cormack, G., Clarke, C., Butt, S. “Reciprocal Rank Fusion outperforms Condorcet and individual Rank Learning Methods.” SIGIR, 2009.
13. Jégou, H., Douze, M., Schmid, C. “Product Quantization for Nearest Neighbor Search.” IEEE TPAMI, 2011.
14. Karpathy, A. “LLM Knowledge Bases.” X post, 2026.

Implementation

15. pgvector: https://github.com/pgvector/pgvector
16. llama.cpp: https://github.com/ggerganov/llama.cpp
17. TurboQuant Pro: https://github.com/ahb-sjsu/turboquant-pro
18. NATS JetStream: https://nats.io

This document consolidates information from ATLAS_AI_README.md, atlas-agi-hpc-implementation-plan.md, start_atlas.sh, the turboquant-pro expansion plan, and empirical benchmarks run on 2026-04-04.